Harshitha Vutukuru Muralikrishna

harshitharaghu22@gmail.com +91 8328543948 Portfolio GitHub LinkedIn

PROFESSIONAL SUMMARY

Machine Learning Engineer with 3 years of experience building **production-scale ML solutions** for clients like **Societe Generale**. Skilled in designing **end-to-end ML pipelines**, developing high impact models and deploying low-latency systems on **AWS SageMaker, Lambda, Docker, Kubernetes**. Strong foundation in **MLOps, model monitoring, NLP and computer vision**. Worked for improving model performance, reducing latency, and enabling reliable cloud deployments.

EDUCATION

Master of Science in Computer Science (Data Science & ML Specialization)
California State University, Los Angeles | *Jan 2023 - Dec 2024*

Bachelor of Engineering in Information Science & Engineering
RNS Institute of Technology, Bengaluru | *Aug 2015 - Jun 2019*

TECHNICAL SKILLS

- Machine Learning and Deep Learning: PyTorch, TensorFlow, Scikit-learn, XGBoost, CNNs
- Cloud and MLOps: AWS SageMaker, Lambda, S3, Kubernetes, Docker, Jenkins, CI/CD pipelines
- Data: Pandas, NumPy, SQL, PySpark
- Programming: Python, MySQL
- Frameworks and Tools: Git, ML flow, Cloud Watch

PROFESSIONAL EXPERIENCE

Accenture - Application Development Analyst

Bengaluru, India | *Sep 2019 - May 2022*

- Built fraud detection models for **Societe Generale**'s global banking systems using **Logistic Regression**, **Random Forest**, and **XGBoost** on 5M transaction records, improving F1-score by about 18% and significantly reducing false positives.
- Engineered **data transformation** and **feature-generation** pipelines that made model training faster and more stable, reducing the training time.
- Containerized multiple ML models with **Docker** and deployed them on **Kubernetes**, which reduced the deployment time and made the systems easier to manage at scale.
- Monitoring dashboards using **AWS SageMaker** and **CloudWatch** to track the metrics like model drift and latency.
- Applied **hyperparameter tuning** and **cross-validation** to improve accuracy and model performance.

PROJECTS

Brain Tumor Detection using Deep CNN's

- Built CNN-based Model for **early detection and classification of brain tumors** from MRI scans, improving diagnostic accuracy and reducing false negatives and false positives.
- Preprocessed and augmented **4,602 MRI images** which includes resizing, normalization, rotation and flipping to enhance model generalization.
- Developed and compared **custom CNN (HALNet1/2, SMVNet, JAPNet)** and **transfer learning models (ResNet50, VGG19)** with convolutional, pooling, batch normalization and dropout layers.
- Optimized training using **binary cross-entropy loss, Adam optimizer, learning rate scheduling and early stopping**; evaluated with **ROC-AUC, confusion matrices, and precision-recall metrics**.
- Used **TensorFlow**, **Keras**, **Scikit-learn**, **Matplotlib/Seaborn**, and **Weights and Biases (wandb)** to build end-to-end ML pipelines, perform hyperparameter tuning, and track experiments for reproducible model development.

AWS SageMaker ML Deployment - House Price Prediction

- Designed and implemented an **end-to-end MLOps pipeline on AWS SageMaker** for house price prediction which includes data ingestion, training, and deployment.
- Performed **data preprocessing, and feature engineering**; versioned datasets in Amazon S3, and ensured reproducibility across training runs.
- Trained an **XGBoost regression model** using SageMaker built-in algorithms and executed **Automatic Model Tuning (HPO)** to optimize hyperparameters, resulting in **15% improvement in prediction accuracy**.
- Deployed the optimized model as **scalable real-time SageMaker endpoint**, enabling low-latency inference and seamless model version updates.
- Connected the SageMaker endpoint with **AWS Lambda** to enable serverless, low-latency inference, ensuring scalable and production ready model.

CERTIFICATIONS

AWS Certified Machine Learning – Specialty